Urban Risk Analytics API (0.1.0)

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tiles

This is an Endpoint to get the road tiles in MVT format with Risk Overlays

Returns a Mapbox Vector Tile (MVT) for roads in the requested slippy-map tile. Risk scoring is optional and controlled by the `crime`, `collisions`, and `userReportedEvents` toggles. When any toggle is enabled, both `startMonth` and `endMonth` are required in `YYYY-MM` format.

PATH PARAMETERS

`z` required	integer (Z) [0 .. 22] Slippy-map zoom level (0-22).
`x` required	integer (X) >= 0 Slippy-map tile X coordinate.
`y` required	integer (Y) >= 0 Slippy-map tile Y coordinate.

QUERY PARAMETERS

`startMonth` >	Startmonth (string) or Startmonth (null) (Startmonth) Inclusive start month for risk scoring in `YYYY-MM` format. Required when any risk toggle is true.

endMonth >	Endmonth (string) or Endmonth (null) (Endmonth) Inclusive end month for risk scoring in `YYYY-MM` format. Required when any risk toggle is true.
crime >	Crime (boolean) or Crime (null) (Crime) Include crime component in risk scoring.
collisions >	Collisions (boolean) or Collisions (null) (Collisions) Include collisions component in risk scoring.
userReportedEvents >	Userreportedevents (boolean) or Userreportedevents (null) (Userreportedevents) Include user-reported events component in risk scoring.

Responses

> **200** Binary Mapbox Vector Tile payload (`application/vnd.mapbox-vector-tile`).

> **400**
Invalid request. Examples: removed `month` or `crimeType`, invalid tile coordinates, invalid month format, or missing month window when risk toggles are enabled.

> **422** Validation Error

> **503** Database dependency unavailable while generating tile.

GET /tiles/roads/{z}/{x}/{y}.mvt

Response samples

400 422 503

Content type
application/json

Example
Removed month parameter

Copy Expand all Collapse all


```
{
    "error": "MONTH_PARAMETER_REMOVED",
    "message": "month has been removed; use startMonth and endMonth instead
  - "details": {
        "field": "month"
    }
}
```


auth

Register a new user

Creates a user account with email and password.

REQUEST BODY SCHEMA: application/json
required

email required	string (Email) Email address for the account.
password required	string (Password) >= 8 characters Plain-text password. Must be at least 8 characters.

Responses

> **200** Registered user profile.

> **422** Validation Error

POST /auth/register

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "student@example.com",
    "password": "StrongPass123"
}
```

Response samples

200 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
  - "user": {
        "id": 0,
        "email": "string",
        "is_admin": true,
        "created_at": "2019-08-24T14:15:22Z"
    }
}
```

Log in with email and password

Authenticates a user and returns a bearer access token.

REQUEST BODY SCHEMA: application/json
required

email required	string (Email) Email address for the account.
password required	string (Password) >= 8 characters Plain-text password. Must be at least 8 characters.

Responses

> 200 Access token and authenticated user profile.

> 422 Validation Error

POST /auth/login

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "student@example.com",
    "password": "StrongPass123"
}
```

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "access_token": "string",
    "token_type": "string",
  - "user": {
        "id": 0,
        "email": "string",
        "is_admin": true,
        "created_at": "2019-08-24T14:15:22Z"
    }
}
```

Get current user

Returns the currently authenticated user from the bearer token.

AUTHORIZATIONS: > *HTTPBearer*

Responses

> **200** Current authenticated user profile.

GET /me

Response samples

200

Content type
application/json

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```
{
  - "user": {
        "id": 0,
        "email": "string",
        "is_admin": true,
        "created_at": "2019-08-24T14:15:22Z"
    }
}
```

Update current user profile

Updates the authenticated user's email and/or password.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json
required

`email` >	Email (string) or Email (null) (Email) New email address. Optional.
`password` >	Password (string) or Password (null) (Password) New plain-text password. Optional, minimum 8 characters.

Responses

> **200** Updated user profile.

> **422** Validation Error

PATCH `/me`

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "new-email@example.com"
}
```

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "user": {
        "id": 0,
        "email": "string",
        "is_admin": true,
        "created_at": "2019-08-24T14:15:22Z"
    }
}
```


reported-events

Create a reported event

Creates a new user-reported event (crime or collision). If a bearer token is provided, the report is linked to the authenticated user; otherwise it is stored as anonymous.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json
required

`event_kind` required	string (Event Kind) Enum: `"crime"` `"collision"` Type of reported event.

event_date required	string <date> (Event Date) Calendar date when the event happened (`YYYY-MM-DD`).
event_time >	Event Time (string) or Event Time (null) (Event Time) Optional local time when the event happened (`HH:MM`).
longitude required	number (Longitude) [-180 .. 180] Event longitude in WGS84.
latitude required	number (Latitude) [-90 .. 90] Event latitude in WGS84.
description >	Description (string) or Description (null) (Description) Optional free-text description from the reporter.
crime >	ReportedCrimePayload (object) or null Required when `event_kind=crime` ; must be omitted for collision events.
collision >	ReportedCollisionPayload (object) or null Required when `event_kind=collision` ; must be omitted for crime events.

Responses

> **201** Reported event created successfully.

— **400** Invalid request payload or validation error.

— **401** Authentication token provided but invalid/expired.

> **422** Validation Error

— **503** Database unavailable.

POST /reported-events

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all


```
{
 - "crime": {
      "crime_type": "Burglary"
   },
   "description": "Suspicious activity near junction after 9pm.",
   "event_date": "2026-03-10",
   "event_kind": "crime",
   "event_time": "21:30",
   "latitude": 53.699,
   "longitude": -1.518
}
```


Response samples

201 **422**

Content type
application/json

Copy Expand all Collapse all


```
{
```

```
  - "report": {
        "id": 1024,
        "event_kind": "crime",
        "reporter_type": "authenticated",
        "month": "2026-03",
        "event_date": "2026-03-10",
        "event_time": "21:30",
        "longitude": -1.518,
        "latitude": 53.699,
        "segment_id": 0,
        "snap_distance_m": 0,
        "description": "string",
        "admin_approved": true,
        "moderation_status": "string",
        "moderation_notes": "string",
        "created_at": "string",
      + "details": { … },
        "user_id": 0,
        "reporter_email": "string",
        "moderated_by": 0,
        "moderated_at": "string"
    }
}
```

List my reported events

Returns a cursor-paginated list of reports submitted by the currently authenticated user. Supports optional moderation status and event-kind filters.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

`limit`	integer (Limit) [1 .. 100] Default: 20 Maximum number of items to return per page.

cursor >	Cursor (string) or Cursor (null) (Cursor) Pagination cursor in `created_at|id` format.
status >	Status (string) or Status (null) (Status) Filter by moderation status: `pending`, `approved`, or `rejected`.
event_kind >	Event Kind (string) or Event Kind (null) (Event Kind) Filter by event kind: `crime` or `collision`.

Responses

> **200** My reports returned successfully.

— **400** Invalid filter/cursor parameters.

— **401** Authentication required.

> **422** Validation Error

— **503** Database unavailable.

GET /reported-events/mine

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "items": [
      + { … }
    ],
```

```
  - "meta": {
        "returned": 0,
        "limit": 0,
        "nextCursor": "string",
        "filters": { }
    }
}
```


List reported events for admin moderation

Admin-only endpoint that returns a cursor-paginated report list with moderation, reporter identity, event kind, and month-range filtering.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

`from` >	From (string) or From (null) (From) Inclusive start month in `YYYY-MM` format.
`to` >	To (string) or To (null) (To) Inclusive end month in `YYYY-MM` format.
`limit`	integer (Limit) [1 .. 200] Default: `50` Maximum number of items to return per page.
`cursor` >	Cursor (string) or Cursor (null) (Cursor) Pagination cursor in `created_at|id` format.
`status` >	Status (string) or Status (null) (Status) Filter by moderation status: `pending`, `approved`, or `rejected`.
`event_kind` >	Event Kind (string) or Event Kind (null) (Event Kind) Filter by event kind: `crime` or `collision`.
`reporter_type` >	Reporter Type (string) or Reporter Type (null) (Reporter Type) Filter by reporter identity type: `anonymous` or `authenticated`.

Responses

> **200** Admin report list returned successfully.

— **400** Invalid filter/month/cursor parameters.

— **401** Authentication required.

— **403** Admin access required.

> **422** Validation Error

— **503** Database unavailable.

GET /admin/reported-events

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "items": [
      + { … }
    ],
  - "meta": {
        "returned": 0,
        "limit": 0,
        "nextCursor": "string",
        "filters": { }
    }
}
```

Moderate a reported event

Admin-only endpoint to approve or reject a reported event and optionally add moderation notes.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`report_id`
required

integer (Report Id)
Unique ID of the reported event to moderate.

REQUEST BODY SCHEMA: application/json
required

`moderation_status`
required

string (Moderation Status)
Enum: `"approved"` `"rejected"`
Admin moderation decision for the report.

`moderation_notes` >

Moderation Notes (string) or Moderation Notes (null) (Moderation Notes)
Optional admin notes explaining the moderation decision.

Responses

> **200** Report moderation applied successfully.

— **400** Invalid moderation payload.

— **401** Authentication required.

— **403** Admin access required.

— **404** Reported event not found.

> **422** Validation Error

503 Database unavailable

— 503 Database unavailable.

PATCH /admin/reported-events/{report_id}/moderation

Request samples

Payload

Content type
application/json

Copy

```
{
    "moderation_notes": "Corroborated with local incident cluster.",
    "moderation_status": "approved"
}
```

Response samples

200 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
```

```
  - "report": {
        "id": 1024,
        "event_kind": "crime",
        "reporter_type": "authenticated",
        "month": "2026-03",
        "event_date": "2026-03-10",
        "event_time": "21:30",
        "longitude": -1.518,
        "latitude": 53.699,
        "segment_id": 0,
        "snap_distance_m": 0,
        "description": "string",
        "admin_approved": true,
        "moderation_status": "string",
        "moderation_notes": "string",
        "created_at": "string",
    + "details": { … },
        "user_id": 0,
        "reporter_email": "string",
        "moderated_by": 0,
        "moderated_at": "string"
    }
}
```

List user-reported events as GeoJSON

Returns user-reported events as a GeoJSON FeatureCollection for map display. Supports moderation, type, month-range, admin-approved, and optional bounding-box filters.

QUERY PARAMETERS

`from` >	From (string) or From (null) (From) Inclusive start month in `YYYY-MM` format.
`to` >	To (string) or To (null) (To)

Inclusive end month in `YYYY-MM` format.

`adminApproved` >
Adminapproved (boolean) or Adminapproved (null) (Adminapproved)
Filter by whether event was admin-approved.

`minLon` >
Minlon (number) or Minlon (null) (Minlon)
Bounding box min longitude.

`minLat` >
Minlat (number) or Minlat (null) (Minlat)
Bounding box min latitude.

`maxLon` >
Maxlon (number) or Maxlon (null) (Maxlon)
Bounding box max longitude.

`maxLat` >
Maxlat (number) or Maxlat (null) (Maxlat)
Bounding box max latitude.

`limit`
integer (Limit) [1 .. 5000]
Default: `500`
Maximum number of features to return.

`status` >
Status (string) or Status (null) (Status)
Filter by moderation status: `pending`, `approved`, or `rejected`.

`event_kind` >
Event Kind (string) or Event Kind (null) (Event Kind)
Filter by event kind: `crime` or `collision`.

`reporter_type` >
Reporter Type (string) or Reporter Type (null) (Reporter Type)
Filter by reporter identity type: `anonymous` or `authenticated`.

Responses

> **200** GeoJSON user events returned successfully.

— **400** Invalid filter/month/bounding-box parameters.

> **422** Validation Error

— **503** Database unavailable.

GET /user-events

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "type": "FeatureCollection",
  - "features": [
     + { … }
    ],
  - "meta": {
        "returned": 0,
        "limit": 0,
        "filters": { }
    }
}
```


watchlists

Read Watchlists

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

watchlist_id > Watchlist Id (integer) or Watchlist Id (null) (Watchlist Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /watchlists

Response samples

200 422

Content type
application/json

Example
WatchlistListResponse

Copy Expand all Collapse all

```
{
  - "items": [
      + { … }
    ]
}
```

Create Watchlist

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

required

name required	string (Name) non-empty
min_lon required	number (Min Lon)
min_lat required	number (Min Lat)
max_lon required	number (Max Lon)
max_lat required	number (Max Lat)
preference >	WatchlistPreferencePayload (object) or null

Responses

> **201** Successful Response

> **422** Validation Error

POST /watchlists

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all

```
{
    "name": "string",
    "min_lon": 0,
```

```
    "min_lat": 0,
    "max_lon": 0,
    "max_lat": 0,
  - "preference": {
        "start_month": "2019-08-24",
        "end_month": "2019-08-24",
     + "crime_types": [ … ],
        "travel_mode": "walk"
    }
}
```


Response samples

201 422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "watchlist": {
        "id": 0,
        "user_id": 0,
        "name": "string",
        "min_lon": 0,
        "min_lat": 0,
        "max_lon": 0,
        "max_lat": 0,
        "created_at": "2019-08-24T14:15:22Z",
     + "preference": { … }
    }
}
```


Update Watchlist

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

watchlist_id required	integer (Watchlist Id)

REQUEST BODY SCHEMA: application/json
required

name >	Name (string) or Name (null) (Name)
min_lon >	Min Lon (number) or Min Lon (null) (Min Lon)
min_lat >	Min Lat (number) or Min Lat (null) (Min Lat)
max_lon >	Max Lon (number) or Max Lon (null) (Max Lon)
max_lat >	Max Lat (number) or Max Lat (null) (Max Lat)
preference >	WatchlistPreferencePayload (object) or null

Responses

> **200** Successful Response

> **422** Validation Error

PATCH /watchlists/{watchlist_id}

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all

Copy Expand all Collapse all


```
{
    "name": "string",
    "min_lon": 0,
    "min_lat": 0,
    "max_lon": 0,
    "max_lat": 0,
  - "preference": {
        "start_month": "2019-08-24",
        "end_month": "2019-08-24",
    + "crime_types": [ … ],
        "travel_mode": "walk"
    }
}
```


Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "watchlist": {
        "id": 0,
        "user_id": 0,
        "name": "string",
        "min_lon": 0,
        "min_lat": 0,
        "max_lon": 0,
        "max_lat": 0,
        "created_at": "2019-08-24T14:15:22Z",
    + "preference": { … }
    }
}
```

Delete Watchlist

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id)

Responses

> **200** Successful Response

> **422** Validation Error

DELETE `/watchlists/{watchlist_id}`

Response samples

200 422

Content type
application/json

Copy

```
{
    "deleted": true,
    "watchlist_id": 0
}
```

watchlist-analytics

Compute Watchlist Risk Score

Runs the watchlist risk algorithm using the watchlist bbox and stored preference window (start_month/end_month). The algorithm builds three core signals inside the bbox: (1) a crime component using crime-type harm weights, monthly recency decay, area normalization, and persistence across active months; (2) a collision component using severity-weighted collision points with recency decay normalized by effective road length; and (3) a lightweight user-report support signal with cluster caps and faster recency decay. Signals are blended with mode-specific weights (walk/drive) into a raw score, then mapped to a normalized 0-100 risk score. The run is persisted, then compared against prior runs with the same signature (month window + crime types + mode). If historical cohort size is below threshold, the service falls back to nearest reference bboxes and compares against cached or recalculated reference runs. Returns a condensed payload with risk result and comparison proof fields.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id`
required

integer (Watchlist Id) > 0
Example: 42
Unique watchlist identifier owned by the authenticated user.

Responses

> **200** Risk score computed and persisted successfully.

— **400** Watchlist is missing required analytics preference fields or has invalid values.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

POST /watchlists/{watchlist_id}/analytics/risk-score

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "watchlist_id": 42,
  - "risk_result": {
        "risk_score": 63,
        "raw_score": 1.6098,
    + "components": { … }
    },
  - "comparison": {
        "cohort_type": "reference_bboxes",
        "cohort_size": 2,
        "subject_score": 63,
        "rank": 3,
        "rank_out_of": 2,
        "percentile": 0,
    + "distribution": { … },
        "sample_size": 2,
        "historical_count": 1,
        "threshold": 2,
    + "reference_ids": [ … ]
    }
}
```


List Previous Watchlist Risk Runs

Returns previously persisted analytical risk-score runs for the selected watchlist, ordered newest first. Includes risk score, component breakdown, and stored comparison metadata for each historical run.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id) > 0 Example: `42` Unique watchlist identifier owned by the authenticated user.

QUERY PARAMETERS

`limit`	integer (Limit) [1 .. 500] Default: `50` Example: `limit=50` Maximum number of historical runs to return.

Responses

> **200** Historical risk runs returned successfully.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

GET `/watchlists/{watchlist_id}/analytics/risk-score/runs`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "watchlist_id": 42,
  - "items": [
     + { … }
    ]
}
```


Forecast Watchlist Next Month

Computes next-month forecast for a watchlist using the watchlist bbox plus historical monthly baseline from startMonth to the last complete month. The forecast blends official crime counts, approved user-reported crime signal, and collision severity points, then applies recency weights (lambda=0.85) and mode-based emphasis (walk/drive). Returns score, conservative band, expected crime/collision outputs, and Poisson-style intervals.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id) > 0 Example: 2 Unique watchlist identifier owned by the authenticated user.

REQUEST BODY SCHEMA: application/json
required

`startMonth` required	string (Startmonth) Historical baseline start month in YYYY-MM format.
`mode`	string (Mode) Default: `"walk"` Travel mode emphasis. Supported values: walk or drive (aliases accepted).

`crimeTypes` Array of strings (Crimetypes)
Optional crime type filter. Empty list means all crime types.

Responses

> **200** Forecast computed successfully.

— **400** Invalid startMonth/mode or invalid watchlist bbox configuration.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

POST `/watchlists/{watchlist_id}/analytics/forecast`

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all

```
{
  - "crimeTypes": [
        "Burglary",
        "Robbery"
    ],
    "mode": "drive",
    "startMonth": "2025-01"
}
```

Response samples

200 422

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "generated_at": "2026-03-14T20:44:54Z",
  - "forecast": {
        "score": 100,
        "band": "red",
        "expected_crime_count": 56,
        "expected_collision_count": 0,
        "expected_collision_points": 0.4037,
    + "intervals": { … },
    + "components": { … }
    }
}
```

Read Root

Responses

> **200** Successful Response

GET /

Response samples

200

Content type
application/json

Copy

```
null
```

Health Check

Responses

> **200** Successful Response

GET /health

Response samples

200

Content type
application/json

Copy

```
null
```